

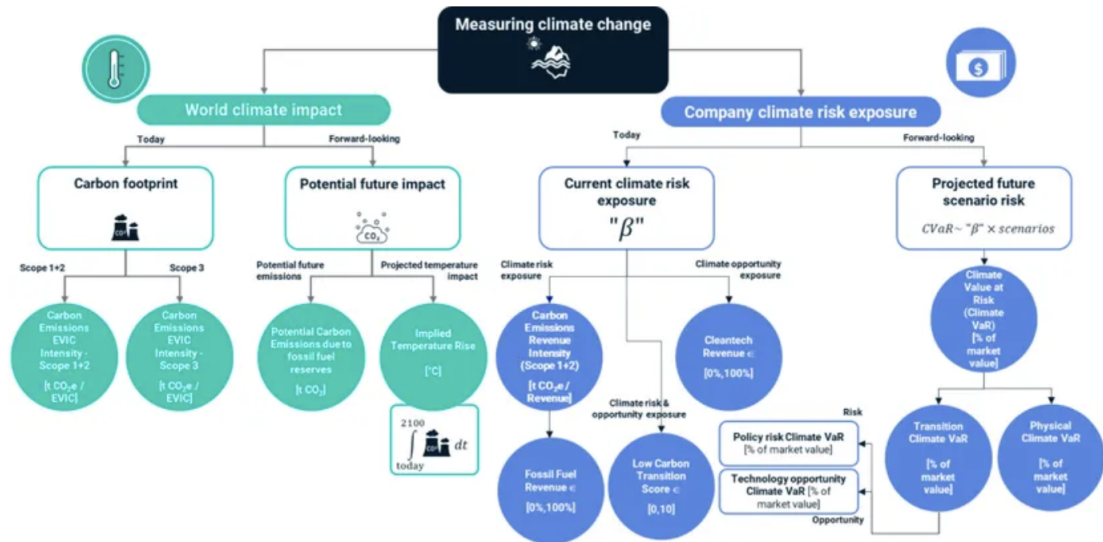
Sustainability aware asset management

Lecture 8: Hedging Carbon Risks

Eric Jondeau

Faculté des HEC, Université de Lausanne
Master of Science in Finance

Objectives of the Lecture



Source: www.msci.com/blog-posts/how-to-evaluate-climate-metrics/03644475407

Objectives of the Lecture

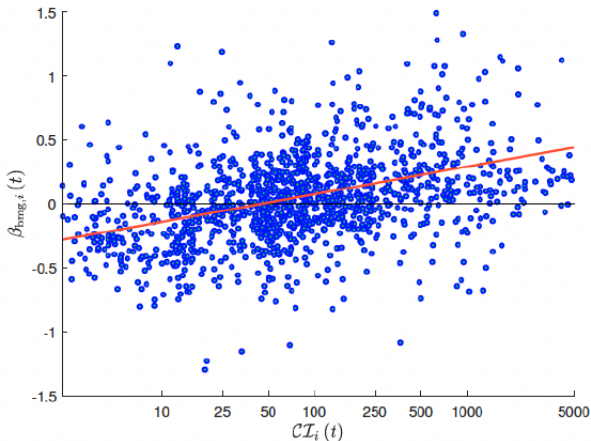
The approach in the previous lecture assumes that climate risk of a company is measured by its current carbon intensity.

But the market perception of the exposure of the company to climate risk may be different.

In this lecture, we define a market measure of carbon, which depends on several dimensions: sector, country, etc.

Market-based vs fundamental-based measures of carbon risk.

Market-based vs fundamental-based measures of carbon risk



Bolton P., and M. Kacperczyk (2021), *Do investors care about carbon risk?*, *Journal of Financial Economics*, 142(2), 517-549

Görge, M., A. Jacob, M. Nerlinger, R. Riordan, M. Rohleder, and M. Wilkens (2019), *Carbon risk*, https://papers.ssrn.com/sol3/papers.cfm?abstract_id=2930897

Roncalli, T., T. Le Guenedal, F. Lepetit, T. Roncalli, and T. Sekine (2020), *Measuring and Managing Carbon Risk in Investment Portfolios*
https://papers.ssrn.com/sol3/papers.cfm?abstract_id=3681266

Objectives of the Lecture

- **Market Measure of Carbon Risk**
- Building a Portfolio Hedged against Climate Risk
- Case Study: Swiss National Bank's Portfolio

Market Measure of Carbon Risk

Three-step approach:

- Define a brown–green score (BGS) for each stock (scoring model)
- Build a brown minus green factor (Fama–French approach)
- Estimate the carbon betas with respect to the BMG factor (multi-factor regression)

Carbon beta = **market** measure of carbon risk

≠

Carbon intensity = **fundamental** measure of carbon risk

Market perception of carbon risk depends on several dimensions: sector, country . . .

Systematic carbon risk

Common risk

Carbon beta

Market measure (general carbon risk exposure, e.g., market repricing risk)

Idiosyncratic carbon risk

Specific risk

Carbon intensity

Fundamental measure (specific carbon risk exposure, e.g., reputational risk)

Görge, M., A. Jacob, M. Nerlinger, R. Riordan, M. Rohleder, and M. Wilkens (2019).
Results based on **Roncalli, Le Guenedal, Lepetit, Roncalli, and Sekine (2020)**

Carbon risk management (Carima) project:

- develops a quantitative tool in order to assess the opportunities of profits and the risks of losses that occur from the transition process
- combines a fundamental approach (carbon score based on fundamentals) and a market-based approach (carbon risk based on market prices)

Step 1: Construction of the Brown – Green score

Step 2: Construction of the BMG factor

Step 3: Multi-factor analysis

Step 1: Construction of the Brown Green Score

Use 55 proxy variables from 4 ESG databases (CDP, Refinitiv, Sustainalytics, MSCI) to define 3 broad indicators:

- **Value chain** ($VC_{i,t}$) captures current emissions related to the production of goods and services – based on Carbon emissions
- **Public perception** ($PP_{i,t}$) represents how the public views a firm with respect to carbon emissions – based on Environmental Score / Performance Band. . .
- **Non-adaptability** ($NA_{i,t}$) is related to the ability of firms to transition from a brown to a green economy – based on Environ. Innovation / Carbon emissions / Preparedness. . .

Each score has a range between 0 and 1. The higher the score, the browner the firm.

A Brown Green Score (BGS) is created for each stock:

$$BGS_{i,t} = \frac{2 + NA_{i,t}}{3} (0.70 VC_{i,t} + 0.30 PP_{i,t})$$

Step 2: Construction of the BMG Factor – Fama-French Approach

Take almost 1,650 firms. They are assigned to one of two mimicking (Brown and Green) portfolios, used to compute the BMG Factor return

Form 2 subsamples with equal size, based on market cap (Small/Big)

For each subsample, form 3 subsamples with equal size, based on a Brown Green Score (Green/Neutral/Brown)

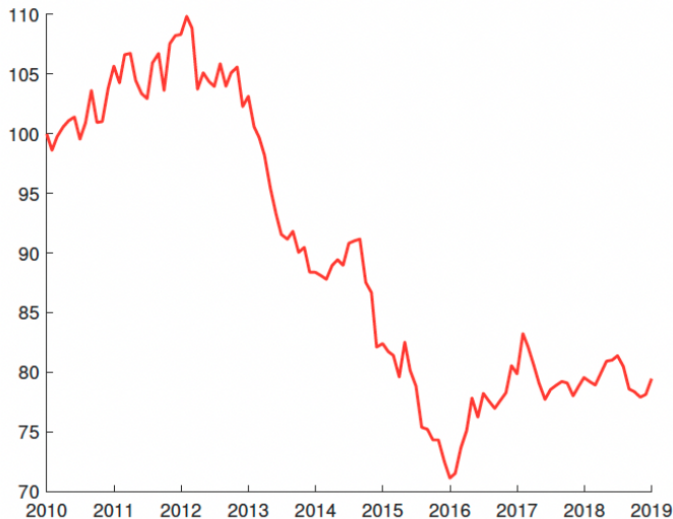
	Green	Neutral	Brown
Small	SG	SN	SB
Big	BG	BN	BB

The “Brown-Minus-Green” (BMG) Factor return $R_{bmg,t}$ is obtained as:

$$R_{bmg,t} = \frac{1}{2} (R_{SB,t} + R_{BB,t}) - \frac{1}{2} (R_{SG,t} + R_{BG,t})$$

where $R_{SB,t}$ (small brown portfolio) is value-weighted by the market cap.

Cumulative Performance of the BMG Factor



Note: Index=100 at end of 2009. Source: Roncalli, Le Guenedal, Lepetit, Roncalli and Sekine (2020).

Step 3: Multi-Factor Analysis

CAPM

$$R_{i,t+1} = a_i + \beta_{m,i}R_{m,t+1} + \varepsilon_{i,t+1}.$$

Fama–French 3-factor model

$$R_{i,t+1} = a_i + \beta_{m,i}R_{m,t+1} + \beta_{smb,i}R_{smb,t+1} + \beta_{hml,i}R_{hml,t+1} + \varepsilon_{i,t+1}.$$

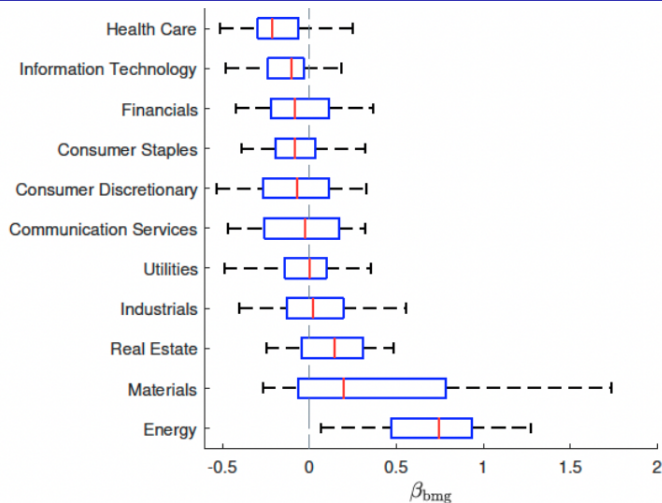
Market + BMG model

$$R_{i,t+1} = a_i + \beta_{m,i}R_{m,t+1} + \beta_{bmg,i}R_{bmg,t+1} + \varepsilon_{i,t+1}.$$

Fama–French 3-factor + BMG

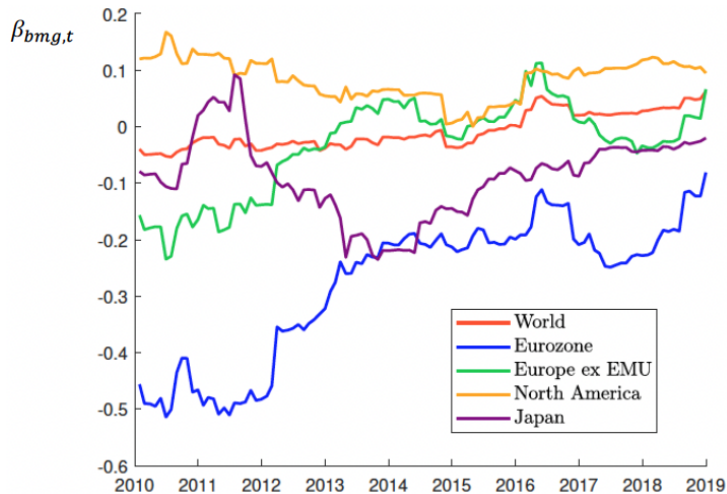
$$R_{i,t+1} = a_i + \beta_{m,i}R_{m,t+1} + \beta_{smb,i}R_{smb,t+1} + \beta_{hml,i}R_{hml,t+1} + \beta_{bmg,i}R_{bmg,t+1} + \varepsilon_{i,t+1}.$$

Carbon Sensitivities across Sectors (Market+BMG Model)



Source: Roncalli, Le Guenedal, Lepetit, Roncalli and Sekine (2020).
Estimation based on stocks from MSCI World from 2010 to 2018.

Dynamics of Relative Carbon Risk by Region



Source: Roncalli, Le Guenedal, Lepetit, Roncalli and Sekine (2020).
Estimation based on stocks from MSCI World from 2010 to 2018.

Advantages and Limits of BMG Approach

Advantages

- Biases in the databases are offset because the BGS scores are based on several databases
- No significant country-specific and sector-specific effects
- No problem of extreme values
- Encompass a lot of climate change-relevant information

Limits

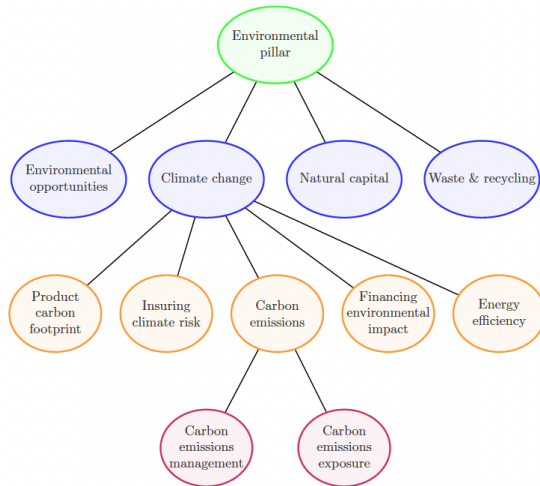
- No differentiation between values near and far from the median of a variable
- No rebalancing schemes
- Correlation between BMG factor and some other factors
- Double counting problems
- Not only carbon risk dimension

Some variables can create more noise than information

What if we consider alternative risk factors: Carbon intensity (Trucost), Carbon emissions exposures (MSCI), Carbon emissions management (MSCI), etc.?

Alternative Risk Factors

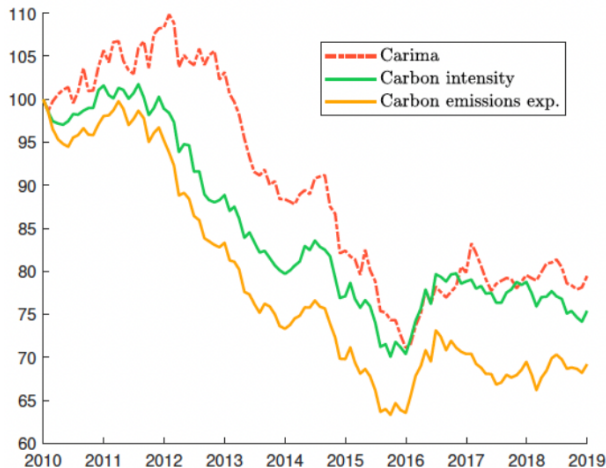
Hierarchy in the environmental pillar (MSCI methodology)



Source: MSCI (2020), Roncalli, Le Guenedal, Lepetit, Roncalli and Sekine (2020).

Alternative Risk Factors

Cumulative performance of the carbon exposure factors



Source: Roncalli, Le Guenedal, Lepetit, Roncalli and Sekine (2020).

Alternative Risk Factors: Comparison of explanatory power of factors

Difference in R^2 between the Market+BMG model and the CAPM

	Full period	First period (2010–2014)	Second period (2014–2018)
CARIMA	1.74	1.16	2.21
Carbon intensity 1–3 (Trucost)	1.77	1.43	2.53
Carbon emissions (MSCI)	2.00	2.18	2.39
Climate change (MSCI)	1.58	1.98	1.83
Environment (MSCI)	1.63	1.35	2.17
Carbon intensity ¹	2.06	1.25	3.13
Carbon emissions ¹	1.91	1.41	2.42

Source: Roncalli, Le Guenedal, Lepetit, Roncalli and Sekine (2020). ¹ The carbon factor is based on the quintile methodology Q5–Q1.

Carbon emissions and intensity are better climate change-related factors to explain stock price fluctuations

Objectives of the Lecture

- Market Measure of Carbon Risk
- **Building a Portfolio Hedged against Climate Risk**
- Case Study: Swiss National Bank's Portfolio

Portfolio Optimization with Carbon Risk – Minimum Variance Portfolio

We consider the Global Minimum Variance (GMV) Portfolio, which solves

$$\begin{aligned} \min_{\{\alpha_t\}} \quad & \frac{1}{2} \alpha_t' \Sigma_t \alpha_t \\ \text{s.t.} \quad & \mathbf{e}' \alpha_t = 1 \end{aligned}$$

The solution is:

$$\alpha_t^* = \frac{\Sigma_t^{-1} \mathbf{e}}{\mathbf{e}' \Sigma_t^{-1} \mathbf{e}}$$

Carbon risk can be introduced

- ① through the covariance matrix Σ_t using a factor model (Σ_t varies over time because $\beta_{m,i}$, σ_m^2 , and $\sigma_{\varepsilon,i}^2$ can vary over time)
- ② through additional constraints on the optimization problem

Portfolio Optimization with Carbon Risk – Covariance Matrix

Assume a factor model, for $i = 1, \dots, N$:

CAPM: $R_{i,t+1} = a_i + \beta_{m,i}R_{m,t+1} + \varepsilon_{i,t+1}$

Market + BMG model: $R_{i,t+1} = a_i + \beta_{m,i}R_{m,t+1} + \beta_{bmg,i}R_{bmg,t+1} + \varepsilon_{i,t+1}$

Denote by σ_m^2 and σ_{bmg}^2 the variances of the market and BMG portfolios. Assume $R_{m,t}$ (long-only) and $R_{bmg,t}$ (long-short) are uncorrelated, and define $\mathbf{D} = \text{diag}(\sigma_{\varepsilon,1}^2, \dots, \sigma_{\varepsilon,N}^2)$

CAPM:

$$\Sigma_t = \beta_m \beta_m' \sigma_m^2 + \mathbf{D} \quad \Leftarrow \text{Only includes market risk in the covariance matrix}$$

Market + BMG:

$$\Sigma_t = \beta_m \beta_m' \sigma_m^2 + \beta_{bmg} \beta_{bmg}' \sigma_{bmg}^2 + \mathbf{D} \quad \Leftarrow \text{Also includes carbon risk in the covariance matrix}$$

Portfolio Optimization with Carbon Risk – Adding Restrictions

Restrict the exposure of the portfolio to (systematic) carbon risk to $\alpha_t \in \Omega$.

- **Limit to portfolio exposure:**

$$\Omega = \left\{ \alpha_t \in \mathbb{R}^N : \beta'_{bmg} \alpha_t \leq \beta^+_{bmg} \right\}.$$

- **Exclusion of stocks with large carbon beta:**

$$\Omega = \left\{ \alpha_t \in \mathbb{R}^N : \alpha_{i,t} = 0 \text{ if } \beta_{bmg,i} > \beta^+_{bmg} \right\}.$$

The program to solve is

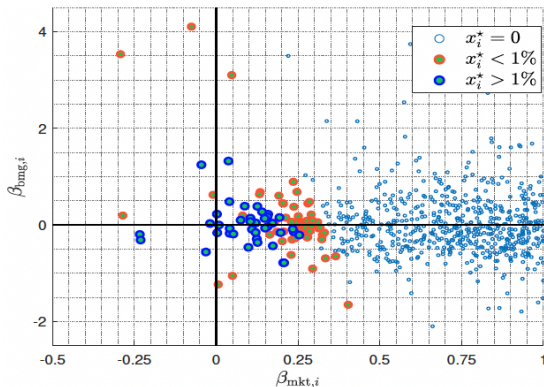
$$\min_{\{\alpha_t\}} \quad \frac{1}{2} \alpha_t' \Sigma_t \alpha_t \quad \text{s.t.} \quad \alpha_t' \mathbf{e} = 1, \quad \alpha_t \in \Omega,$$

where β^+_{bmg} is the maximum tolerance of the investor w.r.t. carbon risk.

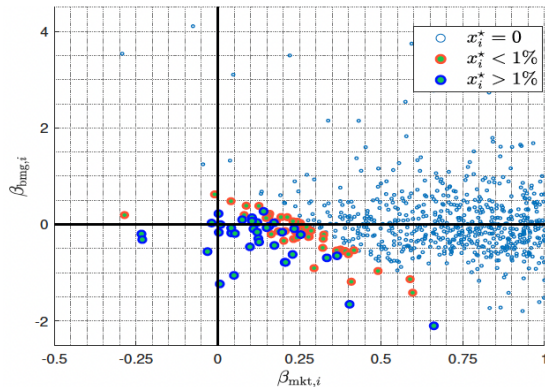
Portfolio Optimization with Carbon Risk

Example based on MSCI World components (end of 2018)

Weights of MV portfolio



Weights of constrained MV portfolio ($\beta_{bmg}^+ = -0.5$)



Source: Roncalli, Le Guenedal, Lepetit, Roncalli and Sekine (2020) ($x_i^* = \alpha_i^*$)

Characteristics of the Portfolio w.r.t. a Benchmark

At date t , we can invest in N risky securities, with returns $\mathbf{R}_{t+1} = (R_{1,t+1}, \dots, R_{N,t+1})'$. A passive investor usually constructs and evaluates its portfolio relative to a benchmark.

Optimal portfolio

Weights: $\alpha_t^{(p)} = (\alpha_{1,t}^{(p)}, \dots, \alpha_{N_t,t}^{(p)})'$

with $\alpha_t^{(p)'} \mathbf{e} = \sum_{i=1}^{N_t} \alpha_{i,t}^{(p)} = 1$

Return: $R_{t+1}^{(p)} = \alpha_t^{(p)'} \mathbf{R}_{t+1} = \sum_{i=1}^{N_t} \alpha_{i,t}^{(p)} R_{i,t+1}$

Benchmark

Weights: $\alpha_t^{(b)} = (\alpha_{1,t}^{(b)}, \dots, \alpha_{N_t,t}^{(b)})'$

with $\alpha_t^{(b)'} \mathbf{e} = \sum_{i=1}^{N_t} \alpha_{i,t}^{(b)} = 1$

Return: $R_{t+1}^{(b)} = \alpha_t^{(b)'} \mathbf{R}_{t+1} = \sum_{i=1}^{N_t} \alpha_{i,t}^{(b)} R_{i,t+1}$

We evaluate the portfolio relative to the benchmark using the tracking error:

$$TE_t^{(p)} = \sqrt{(\alpha_t^{(p)} - \alpha_t^{(b)})' \boldsymbol{\Sigma}_t (\alpha_t^{(p)} - \alpha_t^{(b)})}.$$

Remark: If $\boldsymbol{\Sigma}_t$ is based on monthly returns, the TE should be annualized ($\times \sqrt{12}$).

Enhanced Index Portfolio

For an otherwise passive investor, **enhanced index portfolios** can be constructed in the presence of a benchmark.

The program to solve is

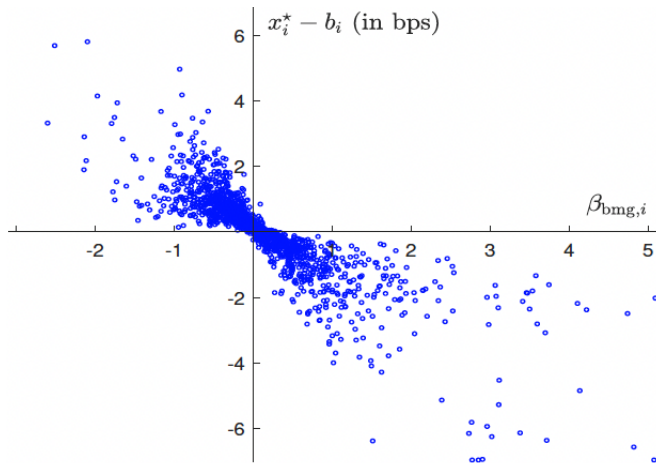
$$\begin{aligned} \min_{\{\alpha_t^{(p)}\}} \quad & \frac{1}{2} \left(\alpha_t^{(p)} - \alpha_t^{(b)} \right)' \Sigma_t \left(\alpha_t^{(p)} - \alpha_t^{(b)} \right) \\ \text{s.t.} \quad & \begin{cases} \alpha_t^{(p)'} \mathbf{e} = 1 \\ \alpha_t^{(p)} \in \Omega \end{cases} \end{aligned}$$

Another constraint may be relative to the benchmark:

$$\Omega = \left\{ \alpha_t^{(p)} \in \mathbb{R}^N : \beta'_{bmg} (\alpha_t^{(p)} - \alpha_t^{(b)}) \leq -\Delta_{bmg} \right\}.$$

Example based on MSCI World index components (end of 2018)

Relationship between $\beta_{bmg,i}$ and $\alpha_{i,t}^{(p)} - \alpha_{i,t}^{(b)}$ (in bps)



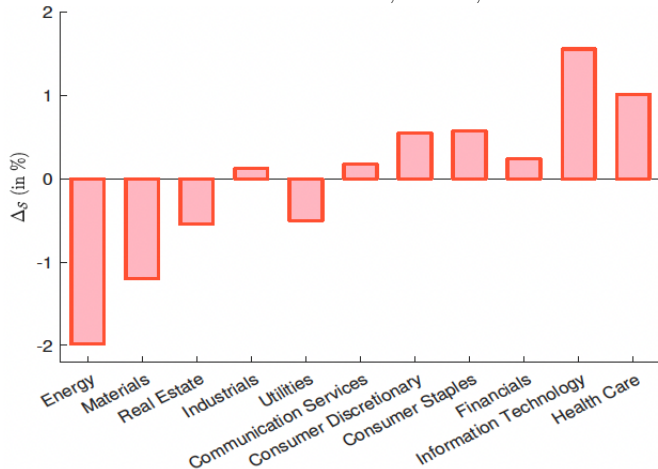
Source: Roncalli, Le Guenedal, Lepetit, Roncalli and Sekine (2020), ($x_i^* = \alpha_i^*$), $\beta_{bmg}^+ = -0.5$.

Sectoral Composition of the Optimal Portfolio

The weight of the energy sector is reduced from 6% in the benchmark to 4% in the optimized portfolio.

The optimized portfolio is less exposed to the sectors that are negatively impacted by an unexpected acceleration in the transition process towards a green economy.

Sectoral exposure $\Delta_s = \sum_{i \in s} (\alpha_{i,t}^{(p)} - \alpha_{i,t}^{(b)})$ (in %)



Source: Roncalli, Le Guenedal, Lepetit, Roncalli and Sekine (2020) ($\beta_{bmg}^+ = -0.5$)

Objectives of the Lecture

- Market Measure of Carbon Risk
- Building a Portfolio Hedged against Climate Risk
- **Case Study: Swiss National Bank's Portfolio**

Based on Fahlenbrach and Jondeau (2022), “Greening the Swiss National Bank's Portfolio”.

Focus on the U.S. equity portfolio of the SNB from 2013 to 2020:

- Capital IQ (Form 13F) for SNB holdings
- SNB foreign exchange reserves $\approx$ CHF 900 bn (end 2020)
- SNB equity portfolio $\approx$ CHF 180 bn ($\sim$ 6,800 stocks)
- Ownership data in Capital IQ: \$132.8 bn (2,767 stocks)
- Carbon data in TruCost: \$126.7 bn (2,064 stocks)

Comparison with BlackRock and Norwegian GPFG.

Strategy rules:

- Rank firms by their Scope 1–2 carbon footprint (emissions/market capitalization)
- Exclude firms with the highest carbon footprint until the cumulative dollar value excluded reaches 1%, 2.5%, or 5% of total market value
- Reinvest proceeds in firms with lowest overall carbon footprint

In 2020, exclusion based on 1%, 2.5%, or 5% corresponds to excluding 42, 87, and 145 firms, respectively.

SNB Portfolio: Excluded Firms

Vistra Corp. (Annual report / Sustainability report) and carbon intensity computation

Vistra Consolidated Financial Results — Year Ended December 31, 2022 Compared to Year Ended December 31, 2021

	Year Ended December 31,	
	2022	2021
Operating revenues	\$ 13,728	\$ 12,077
Fuel, purchased power costs and delivery fees	(10,401)	(9,169)
Operating costs	(1,645)	(1,559)
Depreciation and amortization	(1,596)	(1,753)
Selling, general and administrative expenses	(1,189)	(1,040)
Impairment of long-lived and other assets	(74)	(71)
Operating loss	(1,177)	(1,515)
Other income	117	140
Other deductions	(4)	(16)
Interest expense and related charges	(388)	(384)
Impacts of Tax Receivable Agreement	(128)	53
Loss before income taxes	(1,560)	(1,722)
Income tax benefit	350	458
Net loss	\$ (1,210)	\$ (1,264)



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Vistra corp. Sustainability Report 2022

(GHG emissions in metric tons CO ₂ e)	2019	2020	2021
Scope 1 ¹ emissions	105,523,364	94,290,023	98,749,588
Scope 2 ¹ , location-based emissions	249,068	333,770	242,970
Scope 3 ² emissions	-	2,724,789	2,386,622
Total GHG emissions	105,772,432	97,348,582	101,379,180
Total Generation (MWh)	186,428,605	172,243,738	174,040,471
Emissions Intensity (Scope 1 CO ₂ e/MWh)	0.566	0.547	0.567

$$\text{Carbon intensity} = \frac{98.75 + 0.243}{12077} = 8200$$

SNB Portfolio: Excluded Firms

Excluded company	Sector	Scope 1–2 footprint	Scope 1–2 intensity	Percentage ownership (%)	SNB ptf weight (%)	Contrib. ptf footprint	Contrib. ptf intensity
Vistra Corp.	Utilities	11923.0	8269.1	0.344	0.022	2.57	1.78
SunCoke Energy, Inc.	Materials	7433.9	3207.9	0.226	0.001	0.08	0.03
United States Steel	Materials	5426.2	2800.3	0.185	0.007	0.40	0.21
Cleveland-Cliffs Inc.	Materials	4546.7	6052.4	0.181	0.010	0.46	0.62
Alcoa Corporation	Materials	4151.9	2571.9	0.226	0.010	0.43	0.26
Century Aluminum	Materials	3430.1	3482.5	0.135	0.002	0.06	0.06
NRG Energy, Inc.	Utilities	3169.0	3300.2	0.377	0.028	0.89	0.93
O-I Glass, Inc.	Materials	3146.8	989.8	0.214	0.003	0.10	0.03
AdvanSix Inc.	Materials	2748.0	1881.8	0.227	0.001	0.04	0.03
The AES Corporation	Utilities	2403.7	4473.7	0.376	0.053	1.28	2.39
Covanta Holding Corp.	Industrials	2357.1	2306.0	0.204	0.003	0.07	0.07
Commercial Metals	Materials	2283.1	1426.7	0.225	0.006	0.14	0.09
ALLETE, Inc.	Utilities	2217.5	6238.1	0.225	0.006	0.13	0.36
...	...	...	...	...	...	...	...
Sum					0.819	16.77	30.77

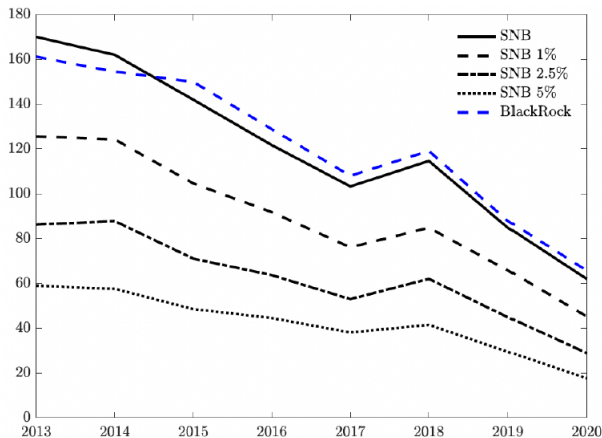
SNB Portfolio – How It Works

- Portfolio footprint in 2020: **62.01 tonnes/mln \$ invested**
- 42 excluded firms: total market value \$1.038 bn (<1% of \$126.7 bn portfolio)
- Excluding these firms reduces footprint by **16.77 tonnes/mln \$ invested**
- Reinvesting in lowest footprint firms increases footprint by **0.01 tonne/mln \$ invested**
- Final footprint: **45.24 tonnes/mln \$ invested**
- Global exclusion changes sectoral balance (utilities/energy/materials excluded; reinvestment into healthcare/IT)

SNB Portfolio

Carbon Footprint (tonnes per \$ mln invested)

Carbon footprint (2013–2020): SNB vs exclusion strategies vs BlackRock.



Strategy rules:

- Rank firms by their Scope 1–2 carbon intensity
- Exclude firms with highest carbon intensity until 1%, 2.5%, or 5% of market value is excluded
- Reinvest proceeds in firms with lowest overall carbon intensity *in the same sector*

In 2020, exclusion based on 1%, 2.5%, or 5% of the total market value corresponds to excluding 42, 87, and 145 firms, respectively

This strategy maintains *sectoral balance*: proceeds reinvested in utilities/energy/materials.

SNB Portfolio: Overweighted Firms

Example of an overweighted firm: Nikola



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SNB Portfolio: Overweighted Firms

Overweighted company	Sector	Scope 1–2 footprint	Scope 1–2 intensity	Percentage ownership (%)	SNB ptf weight (%)	Contrib. ptf footprint	Contrib. ptf intensity
Nikola Corp.	Industrials	0.00	30.6	0.077	0.010	0.00	0.00
Workhorse Group	Cons. discr.	0.03	33.9	0.189	0.001	0.00	0.00
Cadiz Inc.	Utilities	0.06	45.5	0.218	0.001	0.00	0.00
Estée Lauder	Cons. staples	1.25	5.8	0.369	0.001	0.00	0.00
Sunnova Energy	Utilities	1.26	32.4	0.102	0.003	0.00	0.00
Royal Gold	Materials	2.74	37.7	0.226	0.039	0.00	0.01
DMC Global	Energy	5.70	25.2	0.217	0.002	0.00	0.00
AptarGroup	Materials	6.41	18.7	0.224	0.049	0.00	0.01
Cactus	Energy	7.94	36.6	0.224	0.003	0.00	0.00
Dril-Quip	Energy	8.17	28.0	0.224	0.002	0.00	0.00
The York Water	Utilities	11.03	108.7	0.226	0.001	0.00	0.00
NextEra Energy	Utilities	11.83	70.4	0.185	0.008	0.00	0.01
RPC	Energy	12.84	30.6	0.056	0.001	0.00	0.00
...	...	...	...	...	...	...	...
Sum					0.819	0.72	1.89

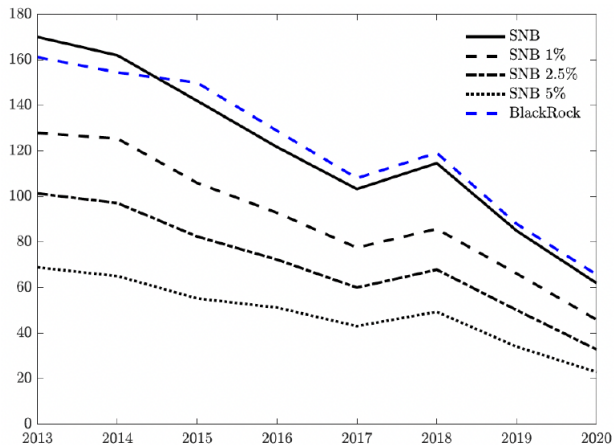
SNB Portfolio – How It Works (sectoral reinvestment)

- We exclude 42 firms with a total market value of \$ 1.038 billion (0.819% of total portfolio value). The portfolio footprint decreases by **16.77 tonnes/mln \$ invested**
- The proceeds of \$ 1.038 billion are reinvested in 41 firms from the same sectors with the lowest carbon footprints. Firms are added until they have the same market value as excluded firms. Added firms increase the portfolio carbon footprint by **0.72 tonnes/mln \$ invested** for a total footprint **45.96 tonnes/mln \$ invested**
- We set the additional weight for each of these 41 firms to be approximately 100% of the original investment. For example, the SNB had an investment of \$ 82.7 million in Edison International. The rebalancing adds \$ 86 million for a total position of \$ 168.7 million
- The additional weight can be restricted further by including more companies with a slightly worse carbon intensity
- While the strategy maintained sectoral balance, it replaced energy utilities with water utilities.

SNB Portfolio

Carbon Footprint (tonnes per \$ mln invested)

Carbon footprint (2013–2020): SNB vs exclusion strategies vs BlackRock.



SNB Portfolio – Financial Performance

	Current SNB portfolio	Global exclusion and reinvestment		Global exclusion and Excl. + sector reinv.	
		1%	2.5%	1%	2.5%
Annual return (%)	15.5	15.5	15.6	15.5	15.6
Annual volatility (%)	15.4	15.5	15.6	15.4	15.4
Sharpe ratio	0.96	0.96	0.95	0.96	0.96
Annual tracking error (%)	1.22	1.15	1.05	1.23	1.21
Carbon emissions (mln tonnes)	6.18	4.64	3.20	4.70	3.61
Carbon footprint (tonnes / mln \$ invested)	120.1	89.8	62.2	91.0	70.5
Carbon intensity (tonnes / mln \$ revenue)	230.3	186.0	126.3	188.2	150.0

The SNB Portfolio and the Exclusion of Coal Firms

In December 2020, Thomas Jordan, Governor of the SNB, announced that in Switzerland a broad consensus has formed in favor of phasing out coal, and that the SNB would exclude firms primarily active in the mining of coal.

In the U.S. equity segment at end-2020, there was not a single firm classified in the “Coal and consumable fuels” category.

In 2019, the SNB invested in 2,263 U.S. firms (with carbon data), total market value \$88.3 bn. This included 109 energy firms (market value of **\$4 billion**, 4.55%). Energy included four coal firms with market value **\$4.5 million** (0.0051%).

Coal firms: Peabody Energy, NACCO Industries, Arch Resources, Consol Energy. Their Scope 1–2 carbon intensity in 2020 is high (2082, 2381, 2625, 7174 tonnes CO₂/\$ million revenue).

Their weight was so small that excluding them has marginal effect: footprint would go from 62 to 61.4 tonnes CO₂/\$ million invested.

Coal: where the footprint really comes from

The 20 U.S. firms that contributed the most to the SNB portfolio carbon footprint in 2020 represent just 1% of the portfolio, with average carbon intensity 4,420 tonnes CO₂/\$ million revenue.

These 20 firms contribute 17 tonnes CO₂/\$ million invested to the SNB footprint (out of 62).

These 20 firms include several electricity power generation and distribution firms, such as Vistra Corp., Allete, and Evergy. These three firms emit between 4800 and 8300 tonnes CO₂/\$ million of revenue because a large portion of their electricity generation comes from burning coal (36% for Vistra, 38% for Evergy, and 61% for Allete).

Key point: The most carbon-intensive firms do not mine coal but burn coal to generate electricity.

End of Lecture 8